Optics & Laser Technology 155 (2022) 108421

Contents lists available at ScienceDirect

Optics and Laser Technology

journal homepage: www.elsevier.com/locate/optlastec

Full length article

Air-slot assisted TM-pass waveguide polarizer based on lithium niobate on insulator

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ARTICLE INFO

Keywords:
Micro-photonic polarizer
Thin-film lithium niobate on insulator (TFLNOI)

ABSTRACT

An air-slot assisted TM-pass waveguide polarizer based on thin-film lithium niobate on insulator (TFLNOI) is proposed. It consists of a segment of ridge-type TFLNOI single-mode strip waveguide and an embedded air-slot. The air-slot has a symmetric tetragonal dipyramid structure consisted of a cuboid in the central part and two identical tetragonal pyramids in two sides, which function as polarization generation and adiabatic approximation to degrade insertion loss, respectively. Waveguide properties and transmission characteristics of the device were simulated using finite difference method. The results show that the waveguide structure with a properly designed geometry supports only a TM mode, while does not the TE mode, which is cut-off in the cuboid air-slot region. Slot length, width and silica substrate thickness all take considerable effects on the insertion loss and extinction ratio. Qualitative and comprehensible explanations are given for the effects. An optimized polarizer has been suggested that exhibits excellent polarization performance, including a larger extinction ratio of 28.4 dB, a lower insertion loss of 0.29 dB, and a broader bandwidth of ~110 nm (1490–1600 nm) with an extinction ratio >20 dB and an insertion loss <0.5 dB. In addition, the proposed device has a length of only 13 μm satisfying on-chip use.

1. Introduction

Silicon-on-insulator (SOI) waveguide enables to implement efficient and ultra-compact on-chip micro-photonic devices due to its high relative refractive index difference (~2.0), ultra-compact waveguide cross-section <1 μm^2 and ultra-small bending radius <10 μm allowed [1,2]. Moreover, because of ultra-compact mode field, light intensity in waveguide increases significantly, causing remarkable increase of material nonlinearity.

Similar waveguide can be realized on the basis of a thin-film $LiNbO_3$ on insulator (TFLNOI) [3–5]. TFLNOI waveguide effectively combines excellent electro-optic, acousto-optic and nonlinear properties of LN crystal and peculiar features of the waveguide, including high relative refractive index difference (~0.7), ultra-compact mode field, small bending radius allowed and increased nonlinearity. The effective combination, together with the possibility of fabricating a low-loss waveguide (<0.1 dB/cm at 1.5 μm [6]), makes the TFLNOI waveguide more potential than the SOI one. So far, a variety of TFLNOI-based photonic devices have been developed, such as frequency converters [7,8], electro-optic or acousto-optic modulators [9,10], frequency comb sources [11,12], photon pair generators [13,14], tunable filters [15,16] and lasers [17,18].

Photonic integrated circuits (PICs) have made significant advances over the last two decades [19]. By combining the potential for ultra-high-speed applications with the availability of large LNOI wafers (6 in.), the PICs based on the TFLNOI are becoming a credible alternative for future optical communications systems whether in the classical or quantum domain [6,20]. However, the high index contrast of the TFLNOI leads to performance degradation of many of these systems due to their strong polarization sensitivity.

Polarization management devices, such as polarization beam splitters [21–24], polarization rotators [25], polarization rotator and splitters [26,27], and polarizers [28–35], are effective solutions to solve the polarization sensitivity problem. Among these devices, polarizer is a basic polarization management device. It is similar to a mode filter that can efficiently output the desired polarization mode while significantly attenuate another orthogonally polarized mode over a short distance by using either special waveguide configurations or highly polarization-dependent materials, such as aluminum zinc oxide [36], graphene [37,38]. Thus, polarizer is a more efficient and cost-effective

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https://doi.org/10.1016/j.optlastec.2022.108421
Received 22 April 2022; Received in revised form 27 June 2022; Accepted 28 June 2022
Available online 4 July 2022

polarization management solution for future optical communication systems that do not require polarization division multiplexing. Recently, people have proposed three main strategies to realize a TFLNOI-based waveguide polarizer on the basis of electro-optic modulation [28], mode leakage [29,30] and hybrid plasma [31,32] as reviewed below.

First, an arbitrary polarization state generator was developed based on electro-optic property of LN [28]. However, the device is as long as 1 cm, and hence does not valid for on-chip use.

Second, based on mode leakage strategy, a polarizer with better polarization performance has been developed [29,30]. The device also suffers from larger size of 1 mm in length.

Third, for the hybrid plasma strategy, although smaller size is possible, e. g., 23 μm [31], 10 μm [32], even ~630 nm [33], the developed devices suffer from either larger insertion loss (>2.0 dB) [31,32] or both larger insertion loss (>1.5 dB) and lower extinction ratio (~14.5 dB) [33].

One can see from preceding reviews that it is challenging to balance the high performance and compact footprint of these previously reported TFLNOI-based waveguide polarizers. It is thus imperative to explore other way towards a TFLNOI-based waveguide polarizer that is of excellent polarization performance and small size valid for on-chip use as well.

Slot waveguide is a peculiar waveguide structure proposed firstly in 2004 [39] and has been used to realize some polarization management devices by use of its polarization selective property. For example, Prakash *et al.* has demonstrated a TE-pass polarizer with an extinction ratio >20 dB on the basis of combined photonic crystal and slot waveguide [34]. However, the device suffers from a higher insertion loss of ~3 dB. Chen *et al.* has reported a TM-pass polarizer implemented using an air slot embedded in a GaAs waveguide on AlAs substrate [35]. However, they failed to consider substrate thickness, which, as shown by present study, affects substantially the device performance. Along with the air-slot waveguide scheme proposed in [35], here we propose an air-slot assisted TM-pass waveguide polarizer consisted of a segment of ridge-type TFLNOI single-mode strip waveguide and an air-slot embedded in it. The air-slot has a symmetric tetragonal dipyramid structure formed by a cuboid in central part and two identical tetragonal pyramids in two sides, which function as polarization generation and adiabatic approximation to reduce insertion loss, respectively. Simulation results show that the device exhibits better polarization performance and two times smaller size than that reported in [35]. In addition, the substrate thickness effect has been independently studied in this work.

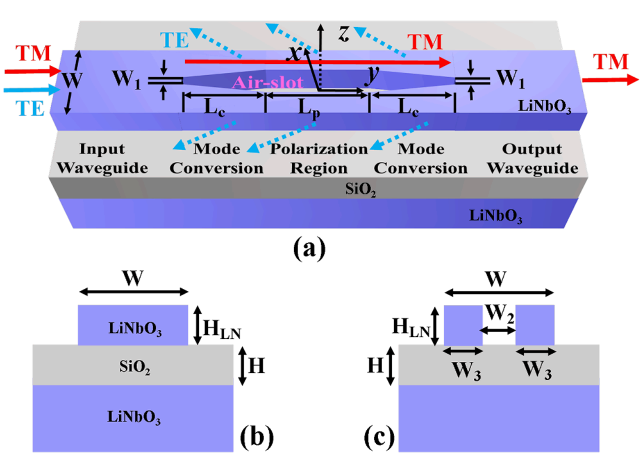

Fig. 1. Structural schematic of proposed air-slot assisted TFLNOI-based TM-pass waveguide polarizer. (a) 3D view, (b) cross-section in I/O regions and (c) cross-section in polarization region.

2. Device structure, principle and numerical method

Fig. 1(a) shows three-dimensional view of structural schematic of proposed air-slot assisted TFLNOI-based TM-pass waveguide polarizer. The device consists of the input/output (I/O) waveguide regions, two mode conversion regions and one polarization region. In the I/O regions, there are two identical ridge-type single-mode channel waveguides with a width W and a height H_{LN} as shown in Fig. 1(b), where the waveguide cross-section in the I/O regions is shown. In the mode conversion and polarization regions, a tetragonal dipyramid air-slot is embedded. In the polarization region, which has a length L_p, the waveguide consists of a cuboid air-slot, which has a width W_2 and a height H_{LN}, and the two LN strip components, which have the same width W_3 and height H_{LN}, as shown in Fig. 1(c), where the waveguide cross-section in the polarization region is shown. The polarization is realized in this region. In the mode conversion regions, the embedded air-slots are same and have a tetragonal pyramid (trapezoid) shape with a length L_c, an initial width W_1 and a final width W_2. The tapered trapezoid air-slot enables to realize adiabatic approximation so as to degrade insertion loss. An *xyz* Cartesian reference frame is fixed at the interface of the silica and LN layers and at the center of the air-slot projected onto the *yz* plane with *x* axis pointing to the depth and *z* axis pointing to the width direction of the strip waveguide. The air-slot is symmetric with respect to all of the three Cartesian coordinate axes *x*, *y* and *z*.

The working principle of the TM-pass polarizer proposed is described below. For a TE wave that is launched into the single-mode waveguide from the input side, it transforms into a slot waveguide mode in the mode conversion region. As the width of the tapered air-slot is increased, the slot waveguide mode broadens in electric field distribution and tends to be cut-off. For a launched TM wave, however, as its electric field direction coincides with the vertical orientation of the air-slot, its electric field distribution is hardly influenced by the air-slot [39]. By designing an appropriate air-slot width W_2, the TM mode can pass through both the mode conversion and polarization regions with a low loss. Instead, as the TE mode operates in a cut-off state, it, as a substrate radiation mode, refracts and leaks into silica substrate layer as indicated by the dotted arrows in Fig. 1(a). As a result, a TM-pass polarizer is achieved. We note from the above-mentioned working principle that the proposed device does not utilize the peculiar properties of LN, such as the electro-optic and nonlinear optical properties, but works on the basis of normal waveguiding property. This implies that the method proposed is applicable to not only the LN waveguide but also the others such as SOI, SiN, and thereby has a larger universality. We note from the above-mentioned working principle that the proposed device does not utilize the peculiar properties of LN, such as the electro-optic and nonlinear optical properties, but works on the basis of normal waveguiding property. This implies that the method proposed is applicable to not only the LN waveguide but also the others such as SOI, SiN, and thereby has a larger universality.

The proposed polarizer is designed on the basis of an X-cut TFLNOI with a thin-film LN device layer thickness H_{LN} = 500 nm and a silica substrate layer thickness H, as indicated in Fig. 1(b). The purpose for choosing the 500 nm thick LN thin film is to ensure strong confinement to the guided wave and be compatible with the thickness of the reported devices, such as frequency conversion device [8] and modulator [10] (the 500 nm is thus called device layer thickness). Here, the reason why the LN instead of the silicon is selected as the handle substrate is that the use of LN can reduce the reflection of substrate radiation mode at the interface between the silica substrate and the LN handle substrate. In addition, as demonstrated below, the thickness of the silica substrate layer affects considerably both the insertion loss and extinction ratio.

Our numerical simulation was carried out in two steps. First, full-vectorial finite-difference frequency-domain (FDFD) approach (commercially available Ansys Lumerical MODE solutions) [40,41] was adopted to model the modal characteristics. Preliminary structural parameters were presented in this step. Second, three-dimensional finite-difference time-domain (3D-FDTD) approach (commercially available Ansys Lumerical FDTD solutions) [41,42] was adopted to further perform a parametric study on the transmission characteristics of the TE or TM mode guided in the polarizer, and optimal geometric parameters were eventually given. To shorten the simulation period and reduce the computational resource consumption, the boundary condition adopted in the 3D-FDTD simulation was chosen according to the symmetries of the structure and mode field. In the case of TM incidence wave, a symmetric boundary condition along the z-axis direction was adopted. In the case of TE incidence wave, however, an anti-symmetric boundary condition was adopted. Material refractive indices of $LiNbO_3$ and silica used in our simulation were taken from [43]. The working wavelength is fixed at 1550 nm unless otherwise specified.

3. Results and discussion

At first, the FDFD mode solver was adopted to carry out an accurate analysis for the modal characteristics of waveguides and the analysis presents preliminary waveguide dimensions. Fig. 2(a) shows effective refractive indices of the TE_{00}, TE_{10}, TM_{00} and TM_{10} modes calculated as a function of the width W of the TFLNOI ridge waveguide. As expected, the mode effective refractive indices increase as the waveguide width W is increased, and higher-order modes appear in the region of W > 1.05 μm. For high-speed optical communication systems, the optical waveguide is usually designed to support only a fundamental mode in order to avoid the signal distortion and thereby performance deterioration caused by the coupling between modes during the transmission. For the TFLNOI waveguide studied here, its single-mode condition is 0.5 μm < W < 1.05 μm for the TE wave and 0.4 μm < W < 1.05 μm for the TM wave.

Confinement factor of a channel waveguide, defined as the ratio of optical power flux in the waveguide region to that in whole space on the plane perpendicular to the waveguide axis, is used to quantify the degree of waveguide constraint on light. It is closely related to the width W of the channel waveguide. Fig. 2(b) shows the confinement factor measured as a function of the width W of the I/O TFLNOI ridge waveguides for the TE_{00}, TE_{10}, TM_{00} and TM_{10} modes. One can see that the confinement factor initially increases with the increased waveguide width W and eventually converges to a constant. It is evident from the definition of confinement factor given above that the decrease of confinement factor is mainly caused by evanescent wave. Thus, for a channel (strip) waveguide with fixed cross-section geometry, the larger the confinement factor is, the smaller the waveguide loss component associated with the evanescent wave is, and hence the smaller the total waveguide loss is. In particular, we note from Fig. 2(a) and 2(b) that the confinement factor in the cut-off region decreases to ~30% for the TE_{00} mode and ~40% for the TM_{00} mode because the evanescent wave is predominant in this case. An overall consideration for single-mode operation, mode field confinement capability and device volume allows us to present a preliminary waveguide width W = 0.9 μm.

The key idea of the suggested TM-pass polarizer is to introduce an air-slot to support TM mode transmission, while make the TE mode cutoff as pointed out above. This requires to design an appropriate slot width W_2. To achieve it, the relationship between effective refractive index of a TE or TM mode guided in the polarization region and the slot width W_2 has been investigated. Fig. 3 shows the calculated air-slot width W_2 dependence of the effective refractive indices of the fundamental TE and TM modes. The two plots clearly show that the effective refractive index decreases, i. e., the mode field distribution broadens with the increased slot width W_2. Comparatively, the effective refractive index of the TE mode decreases more rapidly than that of the TM mode, implying that the TE mode field broadens more rapidly than the TM mode field. We can see that the TE mode is no longer supported as the W_2 is larger than 0.13 μm, whereas the TM mode is always supported as long as W_2 is less than 0.33 μm. This implies that a TM-pass polarizer can be realized as long as the air-slot width W_2 takes a value within 0.13–0.33 μm. As an example, in Fig. 4 we show the field distributions of the fundamental TM mode guided in the I/O waveguide (a) and polarization region (b) in case that the slot width W_2 takes a nearly median value of the data range given above, W_2 = 260 nm. One can see from

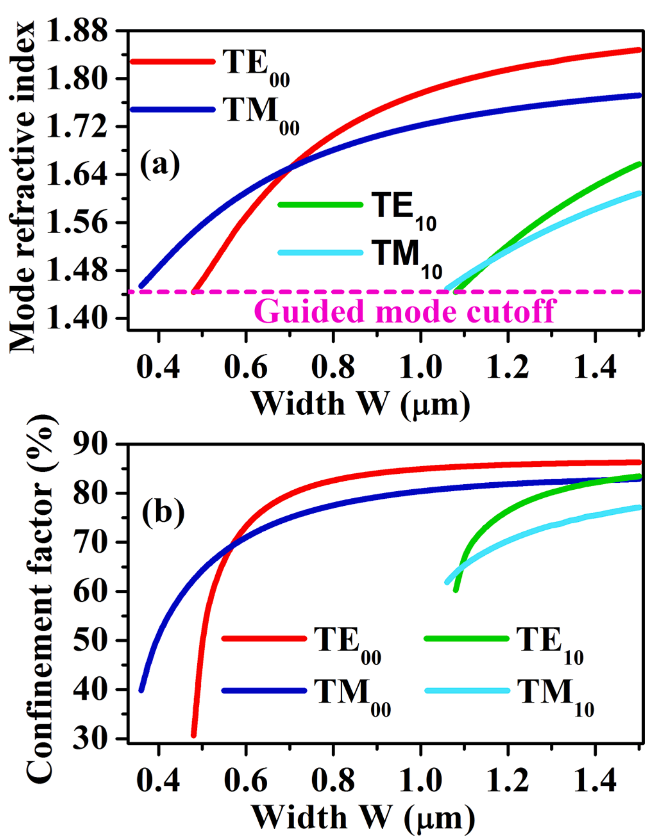

Fig. 2. (a) Effective refractive indices and (b) confinement factors of TE_{00}, TE_{10}, TM_{00} and TM_{10} modes calculated as a function of the width W of the I/O TFLNOI ridge waveguide.

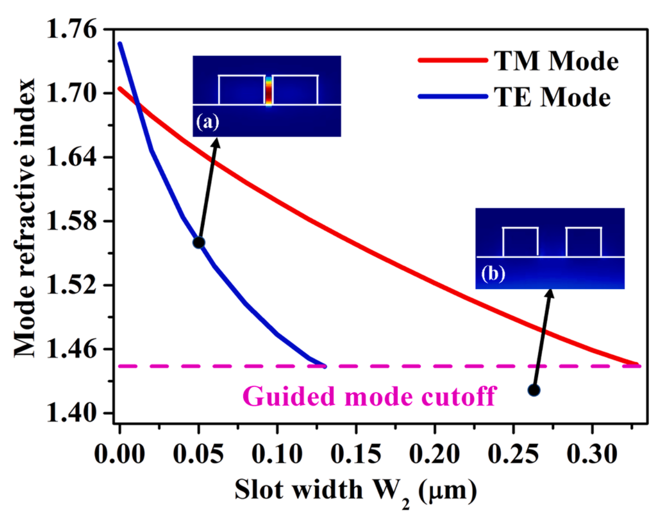

Fig. 3. Air-slot width W_2 dependence of effective refractive indices of the lowest-order TM (red) and TE modes (blue) in the polarization region. Insets show the calculated electric field |E| distribution of a TE mode (a) supported by the slot gap with a width W_2 = 50 nm and (b) not supported by the slot with a width W_2 = 260 nm at 1.55 μm wavelength. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

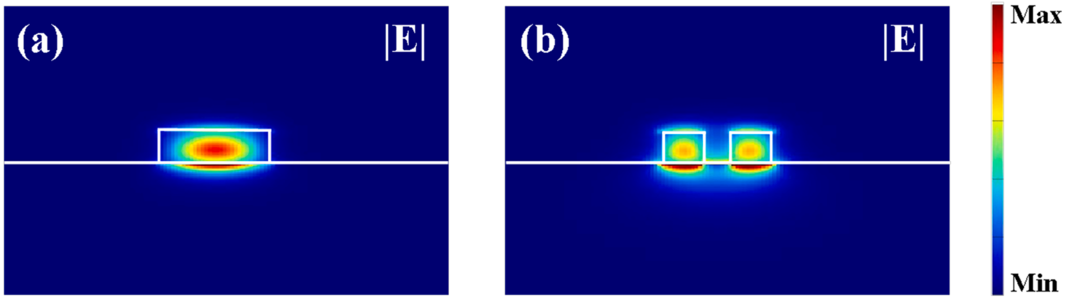

Fig. 4. Electric field |E| distributions of the fundamental TM mode guided in (a) the I/O waveguides and (b) the polarization region in case of $W_2 = 0.26$ μm at 1.55 μm wavelength.

Fig. 4 that the TM mode field is predominantly confined in the high refractive index region of LN layer. In addition to the TM (or TE) mode that is mainly confined in the LN layer, the slot waveguide may also support a gap mode that is confined within the slot. As a representative, two insets in Fig. 3 show the calculated electric field |E| distribution (at 1.55 μm wavelength) of a TE gap mode that is supported by the slot gap with a width $W_2 = 50$ nm (a) while not supported by the slot gap having a width $W_2 = 260$ nm (b).

Due to structural difference, there is a mismatch between the modes guided in the I/O waveguides and polarization region. The mode mismatch degrades the coupling efficiency between the waveguides in the two regions, and hence causes extra insertion loss (coupling loss). To clarify the effect of the air-slot width W_2 on the coupling efficiency, we have calculated the air-slot width W_2 dependence of the coupling efficiency between the fundamental TM modes guided in the I/O and polarization regions using the overlapping integral η, given by [25]:

$$\eta = Re\left[\frac{\left(\int \boldsymbol{E}_1 \times \boldsymbol{H}_2^{*} \cdot ds\right)\left(\int \boldsymbol{E}_2 \times \boldsymbol{H}_1^{*} \cdot ds\right)}{\left(\int \boldsymbol{E}_1 \times \boldsymbol{H}_1^{*} \cdot ds\right)}\right] \frac{1}{Re\left(\int \boldsymbol{E}_2 \times \boldsymbol{H}_2^{*} \cdot ds\right)}, \quad (1)$$

where $\boldsymbol{E}_1$ and $\boldsymbol{H}_1$ denote the electric and magnetic fields in the I/O waveguides, respectively, and $\boldsymbol{E}_2$ and $\boldsymbol{H}_2$ those in the polarization region. The coupling loss arising from the mode mismatch was then calculated according to $-10\log_{10}(\eta)$. The calculated result is shown in Fig. 5(a). One can see that the coupling loss arising from the mode mismatch increases significantly with the increase of air-slot width W_2. The significant increase of coupling loss means a significant decrease in coupling efficiency and a significant increase of insertion loss. To degrade the insertion loss of the device, a tapered air-slot is introduced into the mode conversion region. It satisfies the adiabatic approximation. To facilitate the fabrication, a trapezoidal air-slot structure with an initial bottom width $W_1 = 50$ nm, which has been proved to be feasible in light of fabrication [44,45], was designed as shown in Fig. 1(a). As an example, we have calculated the insertion loss of the TM mode as a function of the length L_c of the mode conversion region in case that $W_1 = 50$ nm, $W_2 = 260$ nm and $L_p = 0$. The result is illustrated in Fig. 5(b), together with the top view of the structure in this case. Here, the insertion loss (*IL*) is defined as [36]

$$IL = -10\log_{10}(T_{TM}), \quad (2)$$

where T_{TM} is the transmittance of the TM mode. The insertion loss is one important parameter used to evaluate the performance of the polarizer. Note that the mode conversions from W_1 to W_2 and from W_2 to W_1 are reciprocal, and the results given in Fig. 5(b) are contributed from the two processes. We can see from Fig. 5(b) that the insertion loss increases slightly with the increased L_c in the initial stage. After passing through a maximum at $L_c \sim 2$ μm, it decreases as the L_c is further increased, and gradually tends to a constant. The feature originates from the competition between the two effects of mode mismatch and adiabatic approximation in the mode conversion region. For shorter L_c, mode mismatch is predominant and the adiabatic approximation effect plays a minor role in the insertion loss in this case. As the L_c is increased, the mode conversion region becomes closer to the ideal adiabatic scenario. In this case, the adiabatic approximation effect plays a predominant role while the mode mismatch effect plays a minor role in the insertion loss. As shown in Fig. 5(b), the effect of L_c on insertion loss is minor and even negligible as the L_c is larger than 5 μm. After a balance of insertion loss and device's footprint, the L_c is chosen as 5 μm here.

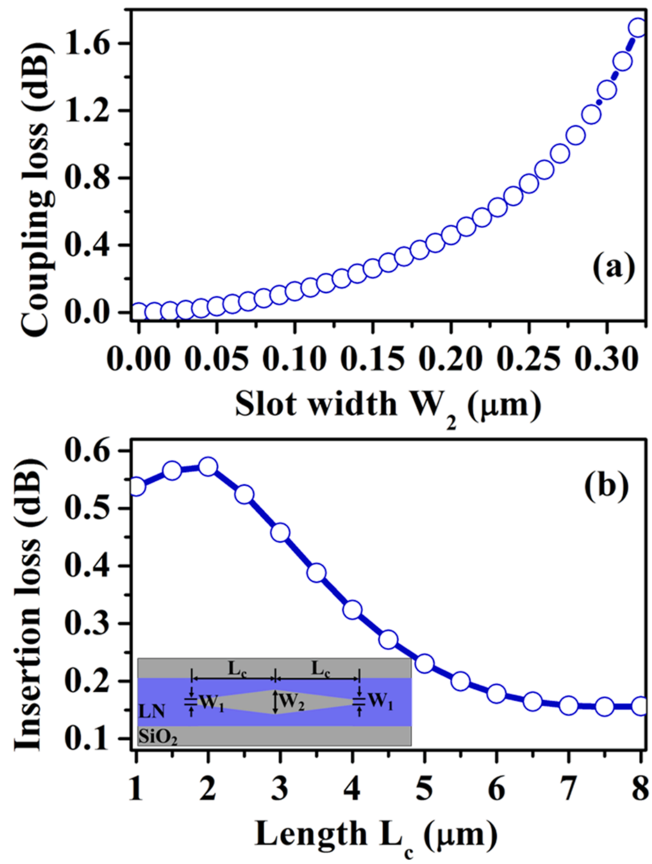

Fig. 5. (a) Air-slot width W_2 dependence of coupling loss arising from TM mode mismatch and (b) mode conversion region length L_c dependence of insertion loss of TM mode in case of $W_2 = 260$ nm, $W_1 = 50$ nm and $L_p = 0$. Inset in (b) shows the top view of the slot structure in case of $L_p = 0$.

The silica is located just beneath the thin-film LN layer. As a dielectric oxide layer, it can prevent energy leakage into the substrate provided it is thick enough. By using the FDFD method, we have calculated the loss due to energy leakage into the substrate as a function of the thickness H of the silica layer for a TM mode guided in both the I/O waveguide and polarization regions. The calculated results in case of $W = 0.9$ μm and $W_2 = 0.26$ μm are shown in Fig. 6. One can see that the

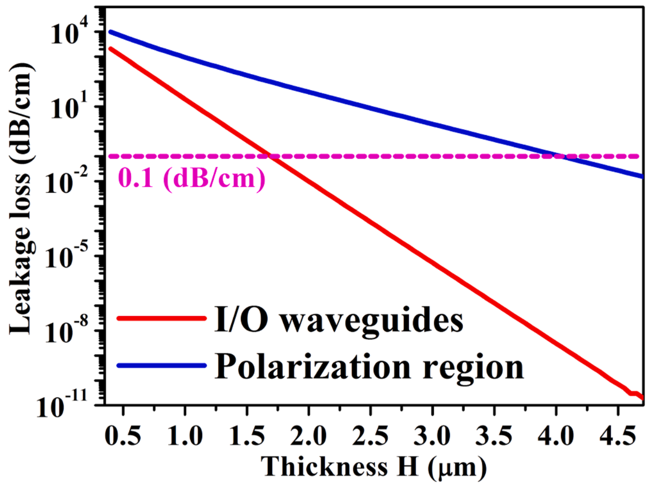

Fig. 6. Silica thickness H dependence of leakage loss of TM mode in the waveguide in the I/O (red) and polarization (blue) regions in the case of W = 0.9 μm and W_2 = 0.26 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

leakage loss reduces as the SiO_2 thickness is increased. It is negligible (<0.1 dB/cm) for the waveguide in the I/O regions as the SiO_2 thickness exceeds 1.7 μm, and for the waveguide in the polarization region as the SiO_2 thickness is greater than 4 μm.

To design a more efficient TM-pass polarizer, we have calculated, using the 3D-FDTD method, the extinction ratio and insertion loss as a function of the polarization region length L_p for five different silica thicknesses of H = 2, 3, 4, 4.7 μm and +∞ (semi-infinite) in case of W = 0.9 μm, W_1 = 0.05 μm and W_2 = 260 nm. The calculated results are shown in Fig. 7. Here, the extinction ratio (*ER*) is defined as [36]

$$ER = 10\log_{10}\left(\frac{T_{TM}}{T_{TE}}\right) \tag{3}$$

where T_{TM} and T_{TE} are the transmittances of TE mode and TM mode, respectively. We can see from Fig. 7(a) that the extinction ratio increases monotonously with the polarization region length L_p in case that the thickness of silica H is semi-infinite. The feature can be explained by the fact that in the polarization region the TE wave continuously refracts and leaks into the silica layer at first and then deeper substrate layer as shown in Fig. 8(a), where evolution characteristics of the dominant electric field component E_z of TE wave on the *xy* plane in the case of H = +∞ (semi-infinite) is illustrated. The continuous energy leakage of TE wave leads therefore to continuous monotonous increase of the extinction ratio with the increase of L_p.

As the silica layer has a finite thickness, however, all of the extinction ratio plots show a non-monotonous relationship to the L_p as shown in Fig. 7(a). Some peaking extinction ratio values appear at some specific lengths L_p. For example, as the silica thickness is H = 2 μm, the maximum extinction ratio appears at L_p = 3 μm, and as the thickness is increased to 4.7 μm, the peaking position moves to L_p = 9 μm. The non-monotonous feature of the extinction ratio can be explained on the basis of propagation characteristics of the substrate radiation mode originating from the cut-off TE wave in the TFLNOI waveguide. As a representative, Fig. 8(b) shows evolution characteristics of the dominant electric field component E_z of TE wave on the *xy* plane in the case of H = 2 μm. It can be seen that the radiated wave of the TE mode is quasi-confined within the silica layer. Different from the case of infinite silica layer thickness, the finite silica layer thickness leads to that part of radiated wave in the silica layer is reflected at the interface between the silica layer and deeper LN substrate, and coupled back into the thin-film LN waveguide, resulting in decrease of the extinction ratio, appearance of some peaking values and thereby non-monotonous feature as shown

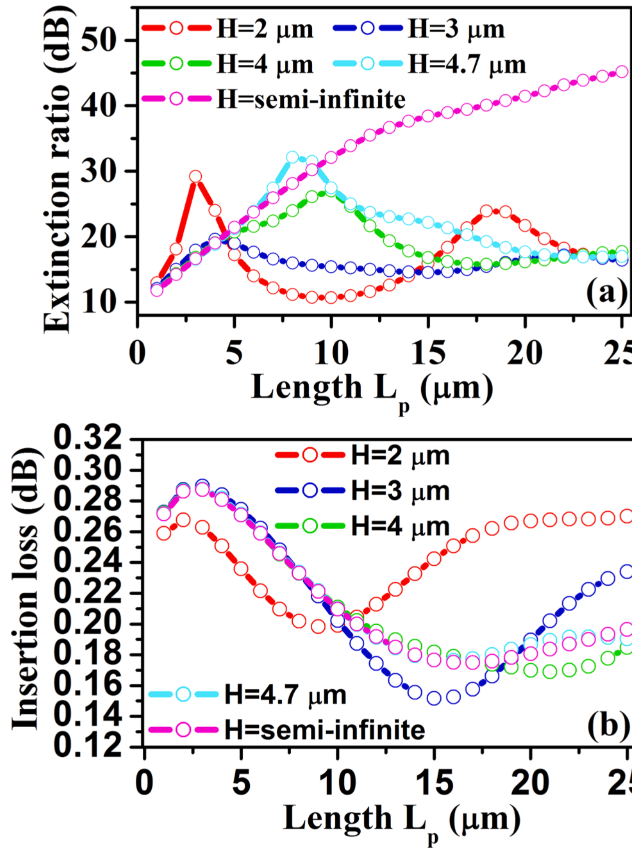

Fig. 7. (a) Extinction ratio and (b) insertion loss of the polarizer device as a function of L_p for different silica thicknesses of H = 2, 3, 4, 4.7 μm and +∞ (semi-infinite) in case of W = 0.9 μm, W_1 = 0.05 μm, and W_2 = 260 nm.

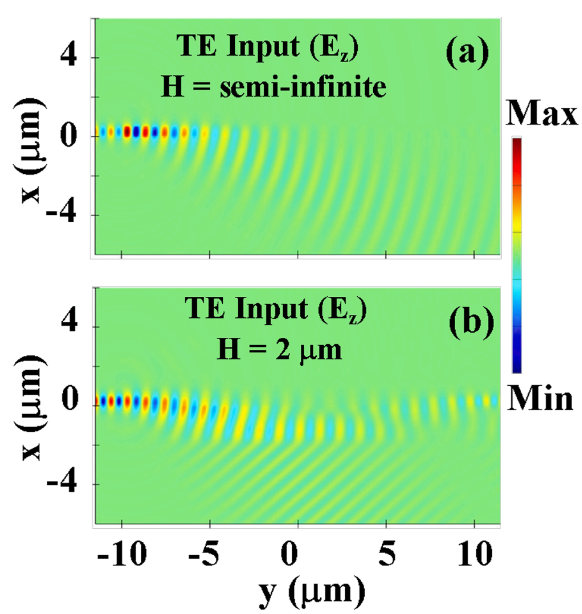

Fig. 8. Evolution characteristics (along the waveguide axis *y*) of dominant electric field component E_z of TE wave on the *xy* plane in the two cases of (a) H = +∞(semi-infinite) and (b) H = 2 μm.

in Fig. 7(a). Due to similar reason of back coupling arising from interface reflection, the insertion loss also has a non-monotonous relationship to the L_p as shown in Fig. 7(b). Nevertheless, the insertion loss keeps always at a lower level of <0.3 dB no matter how the silica layer thickness

H and the polarization region length L_p change in the data ranges studied. Lower insertion loss level is desired for realization of a highly efficient polarizer. As a 0.3 dB insertion loss is tolerable for the application of the device, the optimization for L_p and H only needs to consider the extinction ratio and footprint of the device. A trade-off between them allows us to present optimal L_p and H values, L_p = 3 μm and H = 2 μm, at which the polarizer has an extinction ratio of 29.2 dB and an insertion loss of 0.26 dB. It is worthwhile to point out that, as the length of the proposed device is only 13 μm as stated below, the leakage loss of the TM mode resulting from a thin silica layer is negligible.

It has been shown above that a TM-pass waveguide polarizer can be realized as long as the air-slot width W_2 takes a value within 0.13–0.33 μm. All of the preceding results were obtained in the case that the W_2 optionally takes a nearly median value of the data range, 260 nm, without subjected to an optimization. Next, we try to present optimal W_2 on the basis of the optimal values of W = 0.9 μm, W_1 = 0.05 μm, L_c = 5 μm, L_p = 3 μm and H = 2 μm given above. To achieve it, we have calculated the extinction ratio and insertion loss as a function of the air-slot width W_2 in the case of W = 0.9 μm, W_1 = 0.05 μm, L_c = 5 μm, L_p = 3 μm and H = 2 μm. The calculated results are shown in Fig. 9. One can see that the insertion loss increases slowly at first and then rapidly as the W_2 is increased. A possible reason for the feature is that in the initial stage of the W_2' increase, the extra insertion loss, i. e., the coupling loss arising from the mode mismatch, changes little for the TM mode as shown in Fig. 5(a), while increases definitely for the TE mode, because the TM mode field is influenced less by the air-slot and the TE mode field broadens with the increase of W_2 more rapidly than the TM mode field. As the air-slot width W_2 is increased, the mode mismatch becomes significant for the TM mode, leading to a remarkable increase of the extra insertion loss (coupling loss) and corresponding decline of the extinction ratio and thereby the appearance of the peaking extinction ratio value, which is just located at 260 nm as shown in Fig. 9. In addition, the change of the air-slot width W_2 may affect the back coupling of the TE wave, also leading to the change of the extinction ratio. To achieve a large extinction ratio and a low insertion loss, an optimal air-slot width W_2 = 260 nm is proposed.

In summary, we have obtained optimal structural parameters for the proposed polarizer device. These include W = 0.9 μm, W_1 = 0.05 μm, W_2 = 0.26 μm, H = 2 μm, L_c = 5 μm and L_p = 3 μm. On the basis of these optimal structural parameters we have further calculated the polarizer's performance as a function of working wavelength in the range of 1400–1700 nm. The results are shown in Fig. 10 (see the red plots). We note that the varying features of extinction ratio and insertion loss with the wavelength are quite similar to those with the slot width W_2 as shown in Fig. 9. Both the similarity and varying features may be explained in terms of the effect of the slot width W_2 and working wavelength on the cut-off condition of the TE wave. As shown in Fig. 3, for a given working wavelength, decreasing W_2 tends to increase the mode index, make the TE or TM mode away from the cut-off region, and reduce the loss of the TE or TM wave and thereby the extinction ratio. This is also true for the case that the slot width W_2 is fixed and the operating wavelength is decreased, which makes the TE or TM mode away from the cut-off region.

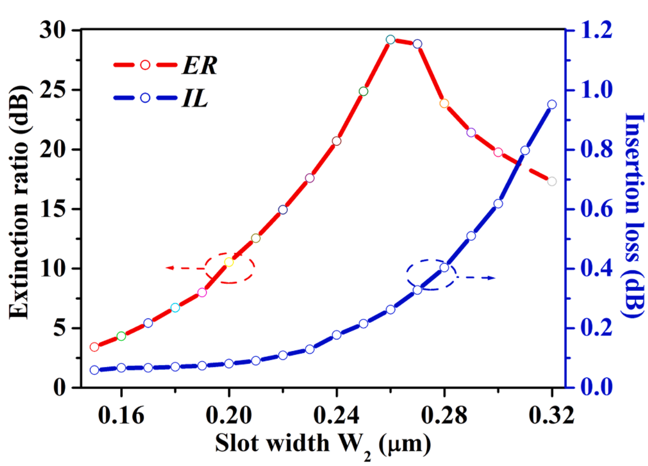

Fig. 9. Extinction ratio and insertion loss calculated as a function of air-slot width W_2 in the polarization region in case of W = 0.9 μm, W_1 = 0.05 μm, L_c = 5 μm, L_p = 3 μm and H = 2 μm.

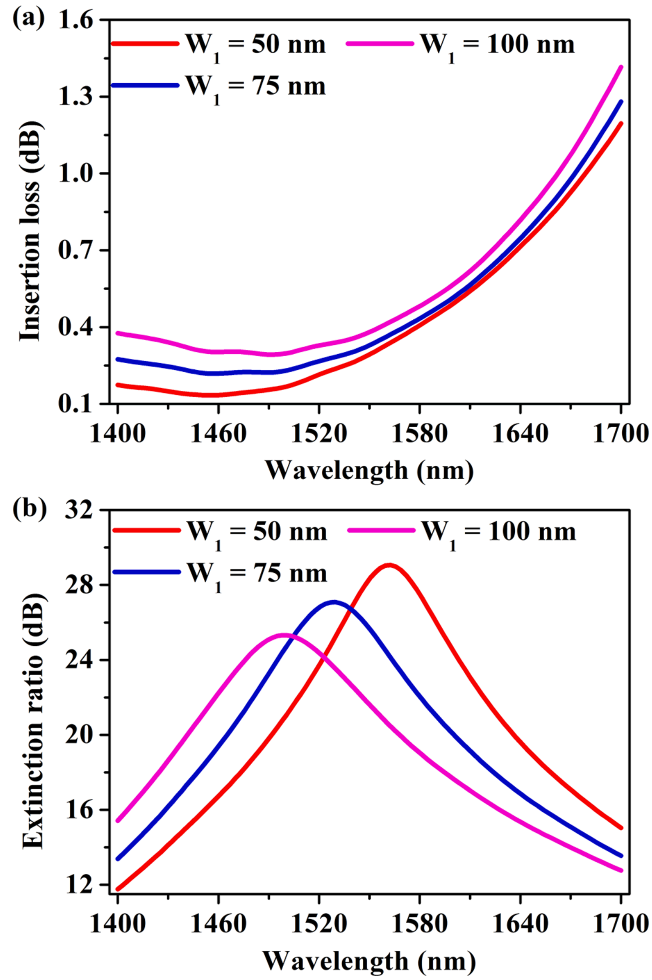

Fig. 10. (a) Insertion loss and (b) extinction ratio of the TM-pass waveguide polarizer calculated as a function of working wavelength for different initial slot widths W_1 = 50, 75 and 100 nm.

Along with the explanation given above, attention is paid to the potential polarization performance of the device. We can see from red plot in Fig. 10(a) that the insertion loss keeps approximately a constant for a wavelength shorter than 1.52 μm and the extinction ratio reaches the maximum of 29 dB at the 1.56 μm wavelength, where the insertion loss is similar to 0.34 dB. In particular, at the working wavelength of 1.55 μm, an extinction ratio of ~28.4 dB and an insertion loss as low as 0.29 dB are expected for the device. It is worthwhile to point out that the values of extinction ratio and insertion loss at 1.55 μm given here are slightly different from those given in Fig. 7. This is due to slightly different refractive indices (at 1.55 μm) adopted in two cases. In the broadband simulation case, we used a refractive index dispersion relation established by carrying out a Cauchy fit to the discrete index values reported in Ref. [43]. In the latter case, the simulation was carried out at just one wavelength of 1.55 μm, and the discrete index value reported in Ref. [43] for this individual wavelength has been used for the simulation, resulting in slight differences of refractive index adopted in two cases and hence of the simulation results.

We also note from the red curve in Fig. 10(b) that the proposed polarizer is expected to have a bandwidth of ~150 nm (from 1490 to 1640 nm) with an extinction ratio of >20 dB. Furthermore, a bandwidth of ~110 nm (from 1490 to 1600 nm) is expected as an insertion loss <0.5 dB is required at the same time (note that the bandwidth 110 nm here is given by the overlapped part of the band given by extinction ratio of >20 dB and that given by insertion loss <0.5 dB).

Considering that the fabrication of 50 nm feature size may be a challenge, we have also estimated the device's performance for wider initial slot width W_1 = 75 and 100 nm. The results are also shown in Fig. 10. We note that the extinction ratio decreases from 28.4 dB to 21.6 dB and the insertion loss increases from 0.29 dB to 0.38 dB at the 1.55 μm wavelength as the W_1 is increased from 50 nm to 100 nm. On the other hand, as the W_1 is increased from 50 nm to 100 nm, the operating bandwidth with an extinction ratio >20 dB and an insertion loss <0.5 dB increases from 110 to 130 nm as a result of blueshift of the extinction ratio plot as shown in Fig. 10(b).

Based on the optimal device geometries, we have simulated the transmission characteristics of the polarizer. Fig. 11 shows the evolution characteristics (along the waveguide axis *y*) of the dominant electric field component E_z of TE wave and E_x of TM wave on the *yz* plane. As expected, the TE wave is in the cut-off state, and, as a substrate radiation mode, refracts and leaks gradually into the silica layer as it propagates along with the waveguide axis as demonstrated above. Instead, the launched TM wave can freely pass through the designed structure almost without experienced an influence in the polarization region.

In comparison with the previously reported hybrid plasma grating-based TFLNOI polarizer [31,32], our device has the advantage of lower insertion loss while maintaining a high extinction ratio. Moreover, present device has a total length of only 13 μm (=$2L_c + L_p$, L_c = 5 μm and L_p = 3 μm), which is two orders of magnitude shorter than that of the TFLNOI polarizer device based on the lateral mode leakage principle [29,30], and meets the demand for on-chip application. In addition, our device also exhibits the advantages over the one with also the air-slot strategy but implemented with GaAs on AlAs [35], which has an extinction ratio of 20.3 dB, an insertion loss of 0.54 dB at the 1.55 μm wavelength, and a device length of 26 μm that are ~28% smaller, two times larger and two times longer than those of our device, respectively.

4. Conclusion

We have demonstrated an air-slot assisted TM-pass TFLNOI waveguide polarizer consisted of a segment of ridge-type TFLNOI single-mode strip waveguide and an embedded air-slot having a symmetric tetragonal dipyramid structure formed by a cuboid in central part and two identical trapezoids in two sides. Simulation shows that, by tailoring the air-slot width, the waveguide structure in the cuboid air-slot (polarization) region supports only a TM mode, while a TE mode in the region is cut-off. As a result, the TM component can freely pass through the structure almost without subjected to an influence while the TE component, as a radiation mode, refracts and leaks into the silica substrate layer, and the TM polarization is realized in the cuboid air-slot region. In the two tetragonal pyramid (mode conversion) regions, the insertion loss of the TM mode is significantly degraded due to the adiabatic approximation effect. Slot length (L_c, L_p), width (W_2) and silica substrate thickness (H) all are key geometric parameters affecting the polarization performance of the device. A balance of polarization performance and size of the device allows us to get optimal structural parameters: W = 0.9 μm, W_1 = 0.05 μm, W_2 = 0.26 μm, H = 2 μm, L_c = 5 μm, L_p = 3 μm and λ = 1.56 μm. The resultant device exhibits excellent polarization performance, including an extinction ratio of 29 dB and an insertion loss of ~0.34 dB. In particular, at the 1.55 μm wavelength, the device has an extinction ratio of ~28.4 dB, an insertion loss as low as 0.29 dB, and a bandwidth of ~110 nm (1490–1600 nm) with an extinction ratio >20 dB and an insertion loss <0.5 dB. In comparison with the previously reported other type polarizers, present device has a lower insertion loss and a smaller size (13 μm in length) valid for on-chip use, while maintaining a large extinction ratio. The device shows great potential for application in various ultra-compact and highly efficient on-chip optical communication systems. The approach can be applied to other waveguides such as SOI, SiN.

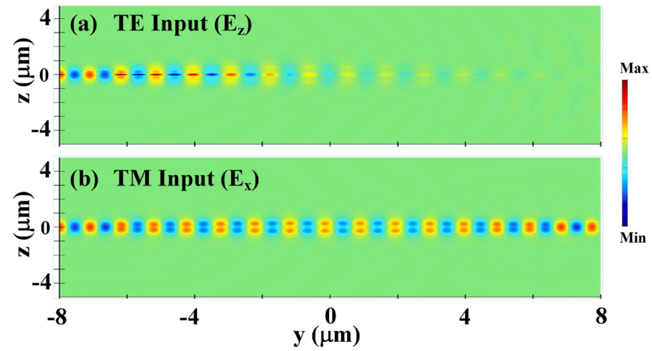

Fig. 11. Transmission characteristics (along the waveguide axis *y*) of dominant electric field component (a) E_z of TE wave and (b) E_x of TM wave on the *yz* plane calculated on the basis of optimal device geometries.

CRediT authorship contribution statement

Jia-Min Liu: Investigation, Software, Writing - Original Draft. **De-Long Zhang:** Conceptualization, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgement

This work was supported by the National Natural Science Foundation of China under Project no. 61875148.

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